

Multiple Choice

1. **Answer:** B

Options: A) Yes, with $p=2$.; B) Yes, with $p=3$.; C) Yes, with $p=5$.; D) No.

Note: The polynomial $x^2 - 12$ satisfies the Eisenstein criterion for irreducibility over $\mathbb{Q}$ with $p=3$ because the coefficients of the polynomial meet the necessary conditions: the constant term (-12) is divisible by 3, the leading coefficient (1) is not divisible by 3, and the coefficient of x (which is 0) is not divisible by 3.

2. **Answer:** D

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: The first statement is false because a nonzero free abelian group of finite rank has a finite number of bases, while the second statement is true as every free abelian group of rank at least 2 can be generated by infinitely many combinations of its basis elements, leading to an infinite number of bases.

3. **Answer:** B

Options: A) If A is true, then B is false.; B) If A is false, then B is false.; C) If A is false, then B is true.; D) Both A and B are true.

Note: If C is false, it means that either both A and B are true or both A and B are false, so if A is false, B must also be false to satisfy the condition of C being false.

4. **Answer:** B

Options: A) 24; B) 30; C) 32; D) 36

Note: The correct answer is 30 because by applying the recursive relation $f(2n) = n^2 + f[2(n-1)]$ systematically starting from $f(0) = 0$, we can compute $f(2)$, $f(4)$, $f(6)$, and ultimately $f(8)$ as 30.

5. **Answer:** A

Options: A) 0; B) 3; C) 12; D) 30

Note: The characteristic of the ring $\mathbb{Z} \times \mathbb{Z}$ is 0 because there is no positive integer n such that n times any element (a, b) in $\mathbb{Z} \times \mathbb{Z}$ equals $(0, 0)$, indicating that the ring is infinite and does not have a finite characteristic.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The statement is true because there exist specific matrices A and B such that their determinants, when added together, equal -1, demonstrating that the sum of two determinants can yield any real number, including -1.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because a polynomial with real coefficients cannot have non-real complex roots unless they occur in conjugate pairs, and the given roots $2 + i$ and $1 - i$ do not form such a pair.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the specific solution of the differential equation must be evaluated at $x=2$ to determine $f(2)$, and without the explicit form of the solution, we cannot conclude that $f(2)$ equals $1/e$.

9. **Answer:** False

Options: A) True; B) False

Note: The given polynomial $x^3 + 2x^2 + 2x + 1$ does not have the roots 1, 4, or 2 in $\mathbb{Z}_7$, as substituting these values does not yield zero, indicating that it cannot be factored into the proposed linear factors in that field.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because an analytic function that maps the complex plane to the real axis must be constant by the open mapping theorem, since non-constant analytic functions cannot have a range that is one-dimensional.

Long Form Response

11. **Answer:** The remainder when $37+49+14$ is divided by 12 is the same as the remainder when the congruences of each number mod 12 are summed and then divided by 12. In other words, since we have $37 \equiv 1 \pmod{12}$

$$49 \equiv 1 \pmod{12}$$

$$14 \equiv 2 \pmod{12} \text{ then therefore, } 37 + 49 + 14 \equiv 1 + 1 + 2 \equiv 4 \pmod{12}.$$

Note: correct because it utilizes the properties of modular arithmetic to simplify the calculation of the total number of eggs modulo 12, allowing for an easier determination of the remainder when the combined total is divided by 12.

12. **Answer:** Let the number of elements in the finite set X be n . The number of subsets containing exactly 2 elements is given by n^2 , and the number of subsets containing exactly 3 elements is n^3 . According to the problem, we have the equation $n^3 - n^2 = 14$. Using the formulas for combinations, we find that $n^2 = \frac{n(n-1)}{2}$ and $n^3 = \frac{n(n-1)(n-2)}{6}$. Substituting these into our equation gives:

$$\frac{n(n-1)(n-2)}{6} - \frac{n(n-1)}{2} = 14.$$

Multiplying through by 6 to eliminate the fractions results in:

$$n(n-1)(n-2) - 3n(n-1) = 84.$$

Simplifying this leads to $n(n-1)(n-2-3) = 84$, or $n(n-1)(n-5) = 84$. Testing various integer values for n , we find that $n = 7$ satisfies the equation. Now, the number of subsets containing exactly 4 elements is given by $n4 = 74 = 35$. Thus, the number of subsets of X that

Note: correct because it effectively sets up the relationship between the combinations of subsets of different sizes using the binomial coefficient formulas, leading to an equation that can be solved to find the value of n and subsequently the number of subsets containing exactly 4 elements.

End of Answer Key